

Day 62 - Software-Defined Networking

Thursday, September 10, 2026

11:16 AM

Software-Defined Networking Info

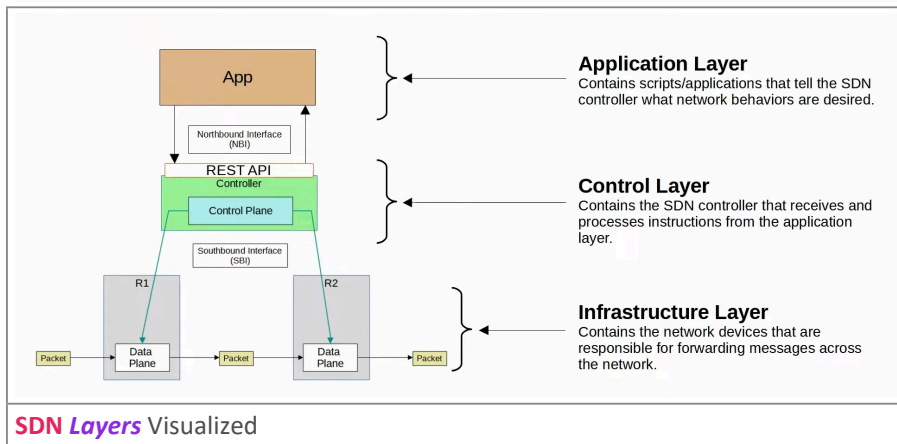
Software-Defined Networking (SDN) is an approach to networking that *centralizes* the *Control plane* into an application called the Controller.

- Traditional *Control planes* use a distributed architecture.
- An **SDN Controller** *centralizes* *Control plane* functions (like calculating routes).
- The **Controller** can interact programmatically with the network devices using **APIs**.
- The **SBI** is used for communications between the controller and the network devices it controls.
- The **NBI** is what allows us to interact with the controller with our scripts and applications.

SDN Architecture

SDN is split into three different *layers*, being:

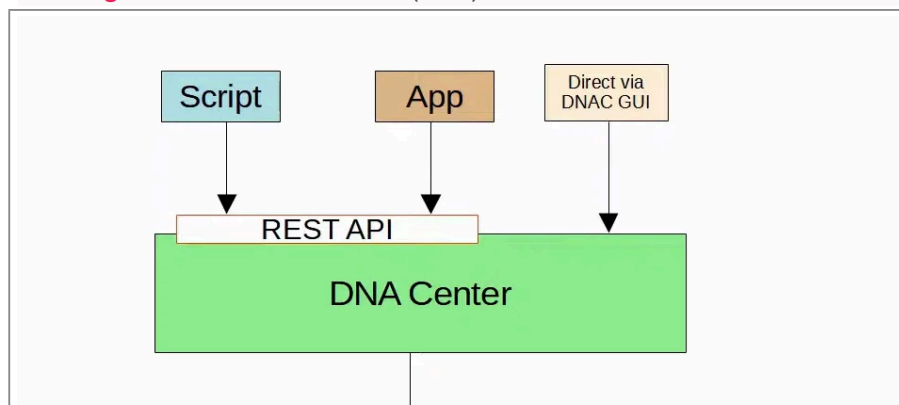
1. *Application Layer*
 - Contains scripts/applications that tell the **SDN Controller** what network behaviors are desired.
2. *Control Layer*
 - Contains the **SDN Controller** that receives and processes instructions from the application layer.
3. *Infrastructure Layer*
 - Contains the network devices that are responsible for forwarding messages across the network.

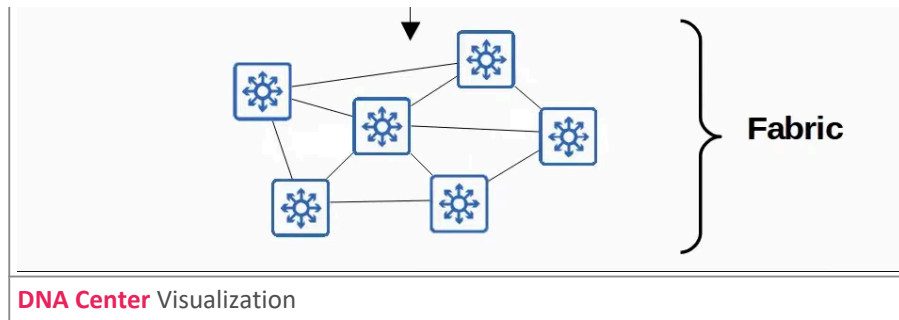


SD-Access

Cisco SD-Access is Cisco's SDN solution for automating campus LANs.

- **Application Centric Infrastructure (ACI)** is their **SDN** solution for automating data center networks.
- **SD-WAN** is their **SDN** solution for automating **WANs**.
- Cisco **Digital Network Architecture (DNA) Center** is the **Controller** at the center of **SD-Access**.





SD-Access Components

- **SD-Access** consists of three different parts called:

1. Underlay

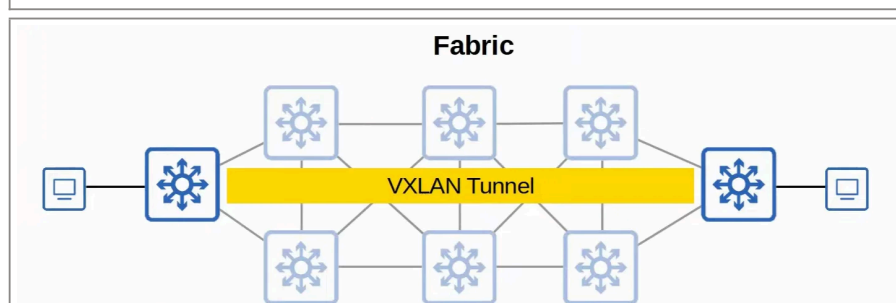
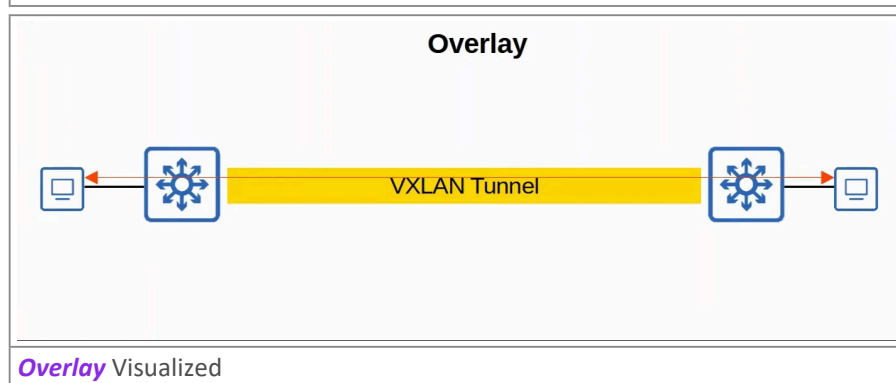
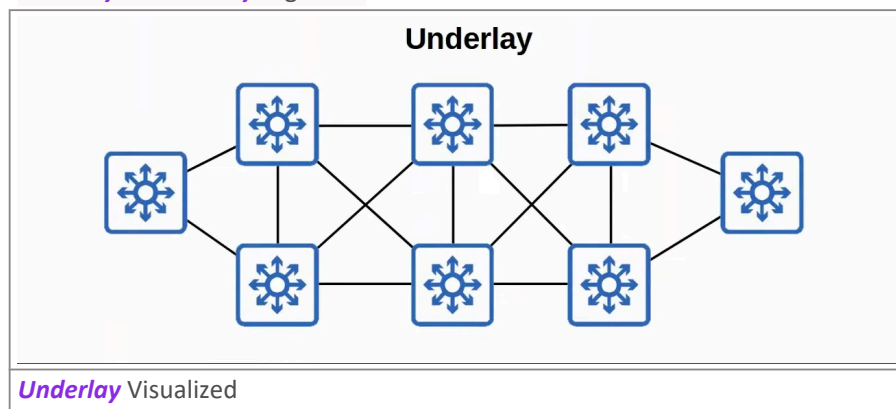
- The **underlay** is the underlying physical network of devices and connections (including **wired** and **wireless**) which provide IP connectivity. (ie. using **IS-IS**)

2. Overlay

- The **overlay** is the virtual network built on top of the physical **underlay** network.
- **SD-Access** uses **Virtual Extensible LAN (VXLAN)** to build tunnels.

3. Fabric

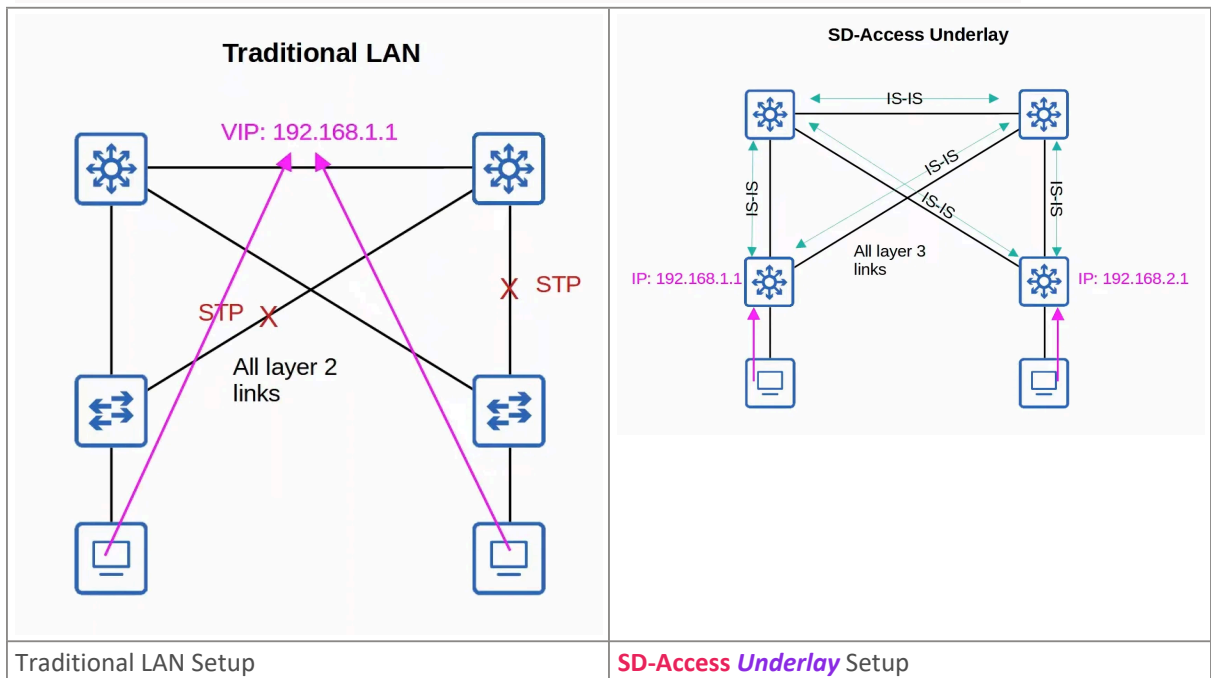
- The **fabric** is the combination of **underlay** and **overlay**; the physical and the virtual network as a whole
- It isn't really considered an actual separate layer, it is just a term that is used to refer to both the **underlay** and **overlay** together.



1. SD-Access Underlay

The **underlay**'s purpose is to support the VXLAN tunnels of the **overlay**.

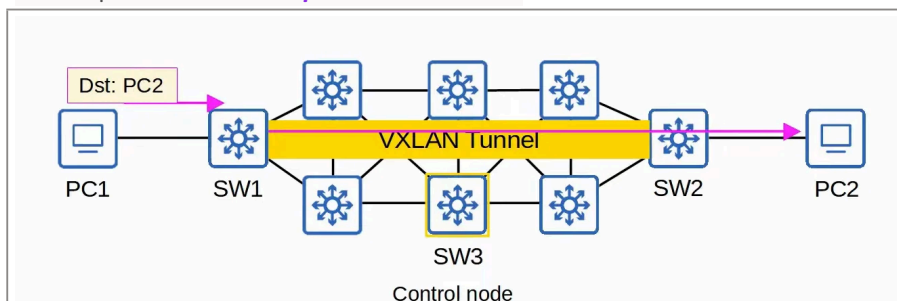
- There are three different roles for switches in SD-Access:
 - Edge nodes**
 - Connect to end hosts.
 - Border nodes**
 - Connect to devices outside of the **SD-Access** domain (ie. **WAN** routers).
 - Control nodes**
 - Use **LISP** (**Locator ID Separation Protocol**) to perform various **Control plane** functions.
- You can add **SD-Access** on top of an existing network (**brownfield** deployment) if your network hardware supports it.
- A new deployment (**greenfield** deployment) will be configured by **DNA Center** to use the optimal **SD-Access underlay**:
 - All switches are **Layer 3** and use **IS-IS** as their dynamic routing protocol.
 - All links between switches are **routed** ports. This means **STP** is not needed.
 - Edge nodes** (**access** switches) act as the default gateway of end hosts (**routed access layer**).



2. SD-Access Overlay

LISP provides the **Control plane** of SD-Access.

- A list of mappings of **Endpoint Identifiers (EIDs)** to **Routing Locators (RLOCs)** is kept.
- EIDs** identify end hosts connected to **edge** switches, and **RLOCs** identify the **edge** switch which can be used to reach the end host.
- Cisco TrustSec (CTS)** provides policy control (**QoS**, security policy, etc.)
- VXLAN** provides the **Data plane** of **SD-Access**.



Cisco DNA Center

Cisco DNA Center has two main roles:

1. The **SDN Controller** in **SD-Access**
2. A network manager in a traditional network (non-**SD-Access**)
 - **DNA Center** is an application installed on Cisco **UCS** server hardware.
 - It has a **REST API** which can be used to interact with **DNA Center**.
 - The **SBI** supports protocols like **NETCONF** and **RESTCONF** (as well as other traditional protocols like **Telnet**, **SSH**, **SNMP**).
 - **DNA Center** enables **Intent-Based Networking (IBN)**.
 - The goal of **IBN** is to allow the engineer to communicate their intent for the network behavior to **DNA Center**, and then **DNA Center** will take care of the details of the actual configurations and policies on devices.
 - Traditional security policies using **ACLs** can become very cumbersome. This is due to multiple reasons:
 - **ACLs** can have thousands of entries.
 - The intent of entries is forgotten with time.
 - Configuring and applying **ACLs** correctly across a network takes a long time and leaves room for error.
 - **DNA Center** allows the engineer to specify the intent of the policy (ie. this group can of users can't communicate with this group, this group can access this server but not that server, etc.), and **DNA Center** will take care of the exact details of implementing the policy.

Traditional Network Management VS DNA Center

1. Traditional Network Management

- Devices are configured one-by-one via **SSH** or **console** connection.
- Devices are manually configured via **console** connection before being deployed.
- Configurations and policies are managed per-device. (distributed)
- New network deployments can take a long time due to the manual labor required.
- Errors and failures are more likely due to increased manual effort.

2. DNA Center-based Network Management

- Devices are centrally managed and monitored from the **DNA Center** GUI or other applications using its **REST API**.
- The administrator communicates their intended network behavior to **DNA Center**, which changes those intentions into configurations on the managed network devices.
- Configurations and policies are centrally managed.
- Software versions are also centrally managed. **DNA Center** can monitor cloud servers for new versions and then update the managed devices.
- New network deployments are much quicker. New devices can automatically receive their configurations from **DNA Center** without manual configuration.